Version of October 4, 2026

MATH 6112. FALL 2026. ASSIGNMENT 2.

DUE: OCTOBER 14, 11:59 PM

Instructions.

(i) You must return your homework typed on letter size paper. Text must be typed in 10pt font or larger.

(ii) The homework must be uploaded in the directory "Assignments" on Canvas by midnight (EDT) of the due day. Submission of a "pdf" file is required.

(iii) Please, keep your arguments short. You do not need to rederive things we have done in class, and can just refer to the class notes. If in doubt, please ask.

(iv) Careful and precise mathematical writing is important. Arguments which are hard to follow will be marked down.

(v) The Homework must reflect your own work. You are welcome to clarify the problems with one another and with me, but in the end you need to work on your own. Use of any source different from the class notes must be explicitly mentioned. This includes use of any AI tool, like `Chatgpt` or the like.

For questions, W 3:30-5, or e-mail, or by appointment.

Homework 2 Problems.

(1) [*15 points.*] This is Exercise 3.2.37 about "Nonorthogonal projections" repeated below for convenience.

Let a projection onto a p-dimensional subspace $\mathcal{V}$ of $\mathbb{C}^n$ be any matrix $B \in \mathbb{C}^{n\times n}$ of rank p such that $B^2 = B$ (idempotency), without requiring symmetry.

(a) Show that a matrix B is a projection if and only if it is similar to $\begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}$.

(b) If V is such that $V^{-1}BV = \begin{bmatrix} I_p & 0 \\ 0 & 0 \end{bmatrix}$, then show that B projects onto the subspace $\mathcal{V}$ spanned by the first p columns of V. This shows that $Bx = x$ for any $x \in \mathcal{V}$.

(c) In $\mathbb{R}^3$, build the orthogonal projection P onto the subspace given by the plane $2x_1 - 3x_2 + x_3 = 0$. Also, build another nonorthogonal projection B onto this same subspace.

(d) In $\mathbb{R}^2$, give an example of a (nonorthogonal) projection B onto one of the coordinate axes, with B of arbitrarily large 2-norm.

(2) [*30 points.*] This exercise introduces you to the concept of *Field of Values* (also called *numerical range* in numerical methods for PDEs).

Let $A \in \mathbb{C}^{n\times n}$. Then, the field of values of A is defined as

$$\mathcal{F}(A) \;:=\; \{x^*Ax \;,\;\; x \in \mathbb{C}^n \;,\;\; x^*x = 1\}\,.$$

Observe that the eigenvalues belong to $\mathcal{F}(A)$, as well as the diagonal entries of A, and that $\mathcal{F}(A)$ is invariant under unitary similarity transformations of A.

An associated concept is that of *numerical radius* of A, denoted as $r(A)$, and defined as:

$$r(A) \;:=\; \max\{\; |z| \;,\;\; z \in \mathcal{F}(A)\;\}\,.$$

(a) Show that if A is normal, then $\mathcal{F}(A) = \mathrm{Co}(\sigma(A))$. That is, the field of values coincide with the convex hull of the eigenvalues of A. [Recall that the convex hull $\mathrm{Co}(S)$ of a finite set S is the set of all convex combinations of the points of S.]

(b) Show that A is Hermitian if and only if $\mathcal{F}(A)$ is a segment of the real line. What are the endpoints?

(c) Give a geometrical description of $\mathcal{F}(\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix})$; in other words, what is it? Also, compute the numerical radius for $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and for A^2.

(d) Show that the numerical radius of a matrix is a norm (nonnegative, positively homogeneous and satisfies the triangular inequality). Is it a consistent matrix norm?

(3) [*15 points.*] This exercise is about determinants and the Cauchy-Binet formula.

The following four properties (i)-(iv) of the determinant can be proven using the definition via the cofactor expansion of the determinant, but you do not need to prove them and can assume them as true.

(i) $\det(cR) = c^n \det(R)$, where $R \in \mathbb{C}^{n\times n}$.

(ii) Interchanging two columns of a matrix multiplies its determinant by -1.

(iii) Adding a multiple of one row to another row does not change the determinant.

(iv) $\det \begin{bmatrix} A & 0 \\ C & B \end{bmatrix} = \det(A)\det(B)$, where $A, B \in \mathbb{C}^{n\times n}$.

Show that the matrix $\begin{bmatrix} A & 0 \\ I & B \end{bmatrix}$ can be taken to the matrix $\begin{bmatrix} AB & 0 \\ B & I \end{bmatrix}$, in the sense that there are block matrices $M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$ and $W = \begin{bmatrix} W_{11} & W_{12} \\ W_{21} & W_{22} \end{bmatrix}$, representing a combination of the above elementary operations (i)-(iii), such that $M \begin{bmatrix} A & 0 \\ I & B \end{bmatrix} W = \begin{bmatrix} AB & 0 \\ B & I \end{bmatrix}$. Note that your task is to give these matrices M and W.

At this point, using (iv) you can infer the Cauchy-Binet identity[1] $\det(AB) = \det(A)\det(B)$.

(4) [*10 points.*] Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and the norm be the 2-norm. Let $A \in \mathbb{F}^{n\times n}$ be invertible, and let $B \in \mathbb{F}^{n\times n}$ be such that

$$\|A - B\| \ < \ 1/\|A^{-1}\| \ .$$

Show that B is invertible.

(5) [*15 points.*] Below, all matrices A, B, C, D are in $\mathbb{C}^{n\times n}$. Show that when $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$, A invertible and $AC = CA$, then $\det(M) = \det(AD - CB)$. Give a counterexample to this equality when $AC \neq CA$.

Challenge [*5 more points.*] Can you show it without assuming A invertible?

(6) [*15 points.*] Exercise 3.2.9 on degree of uniqueness of unitary (orthogonal) transformations.

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[1] Incidentally, the foregoing is probably the simplest proof of the Caucy-Binet identity that I am aware of.